

Adversarially Asymmetric Evidence for Calibrated Fake News Detection

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Course: 22CSP72 — Project Work II

Abstract—Automated fake-news detection scores how an article is written — its tone, its headline, whether it names sources. Every such signal is chosen by whoever wrote the text, so a fabrication composed in newsroom register scores as a genuine report does. On a weighted-sum pipeline of this kind we measured invented articles at 7.3 of 10, with four of six deceptive items presented as credible; the failure was systematic, because when corroboration was absent the verification term withdrew and redistributed its weight onto precisely the signals an author controls. We attribute this to treating adversarial detection as ordinary classification. Where features are adversarially selected rather than drawn from nature, an asymmetry follows, stated here as an admissibility rule: an observation under the author's control may count against an article but never in its favour. We build a scoring model on that rule. Each observation contributes a likelihood ratio estimated from labelled articles rather than a hand-assigned weight; ratios accumulate as additive log-odds into a calibrated probability. Support is admitted only from outside the author's control — corroboration counted in independent sources rather than retrieved articles, and verification of an article's premises against a reference work. Input carrying no verifiable claim is declined rather than scored. On held-out data, accuracy rises from 0.888 to 0.954 against the rule it replaces, precision among flagged articles from 0.848 to 0.975, and calibration error where the system accuses from 0.229 to 0.127.

Index Terms—Adversarial evidence, fake news detection, likelihood ratio, probability calibration, selective prediction, source independence, weight of evidence.

1. INTRODUCTION

The volume of news reaching a reader now exceeds any individual's capacity to verify it. Automated assistance is therefore attractive, and a large body of work addresses it — but almost all of that work is framed as classification: given an article, output "fake" or "real".

Classification is the wrong output for this problem, for three reasons.

A label carries no reason. A reader told that an article is 87% likely to be fake learns nothing they can act on, check, or disagree with. The judgment is unfalsifiable from the reader's position, which is precisely the epistemic situation misinformation already puts them in.

A label hides which evidence it rests on. An article may be untrustworthy because its outlet has a poor record, because its headline oversells its content, because it leans on unattributable authority, or because other outlets report the facts differently.

These are different problems requiring different reader responses, and a single score conflates them.

Confidence is usually dishonest. Guo et al. [6] showed that modern neural classifiers are systematically overconfident: a model reporting 90% confidence may be correct only 60% of the time. Calibration is rarely reported in misinformation systems, so a stated confidence generally cannot be interpreted as a probability.

We take an assessment view instead of a classification view, and separate two questions that a single score conflates. Whether a story holds up is answered by a verdict with its grounds; how well it is written is reported alongside, and never mixed into it. A well-written fabrication should read as exactly that — well written, and unsupported — rather than averaging to something reassuring.

1.1 Contributions

- An admissibility rule for adversarial evidence. Observations under the author's control are admitted only in the incriminating direction, at every stage of the computation including after calibration. Section 7.2 reports what the rule costs and why the cost is worth paying; Section 4.3 states it formally.
- Measured weights in place of assigned ones. Every likelihood ratio is estimated from labelled articles, shrunk to the conservative end of a 95% interval, and stored with the counts that produced it. A factor with too little data to estimate contributes exactly zero and is named as such in the report, rather than receiving an invented weight.
- Correlation-aware combination. The content checks are measured at $r = 0.78$ and amount to 1.2 effective independent factors of four; their summed weight is scaled accordingly, where the previous design counted one piece of evidence four times.
- Independence-aware corroboration. Retrieved coverage is reduced to the sources it actually represents by a curated ownership model, agency-syndication detection and near-duplicate collapse, with every merge reported to the reader. A deterministic relevance gate additionally discards retrieved coverage sharing no anchor with the claim, after the model has spoken.
- Premise verification separate from event verification. A news index cannot adjudicate a claim resting on a false premise, because the invented event is reported nowhere; a reference

work settles it in one lookup (Section 4.4).

- An input admissibility gate. Content carrying no verifiable claim receives no verdict, only a statement of what would make it checkable — a system that answers questions it cannot answer teaches readers to distrust the answers it gets right.
- A leakage-controlled held-out evaluation against the rule this replaces, with calibration reported separately for the accusing and exonerating directions, and an adversarial benchmark that ISOT cannot substitute for.

2. RELATED WORK

Explainable detection. Shu et al. [1] (DEFEND) generate explanations by jointly attending over article sentences and user comments, surfacing the comments most indicative of deception. The explanations are genuine, but the method requires a social-media comment stream, which does not exist for an article a reader has just been sent. Amri et al. [3] (ExFake) combine content with social context and external evidence; again the dependency is on signals that arrive after an article circulates. Our factors are computed from the article and from independent news retrieval only, so a report is available on first sight.

Classification on ISOT. Ahmed et al. [2] established the ISOT dataset and n-gram baselines. Reported accuracy on ISOT is routinely above 0.98, which reflects properties of the corpus more than the difficulty of the task. Verma [14] audits three leakage channels in it: a subject field whose values are disjoint across the two classes, on which a classifier reading no article text at all attains $F1 = 1.000$; a newswire agency tag present in 99.21% of real articles and 0.04% of fake ones; and exact duplicates spanning 19.37% of a naive test split. Removing all three costs only 1.21 F1 points, so a strong lexical signal survives — which is the deeper problem for our purposes, since that signal is exactly what an adversary writing carefully does not emit. We treat high ISOT accuracy as a methodological hazard rather than a result, control the channels we can (Section 6.2), and report the adversarial benchmark (Section 7.7) as the measurement ISOT cannot substitute for.

LLM-based verification. Wang et al. [4] use a large language model to weigh competing crowd reports about a claim. The approach is powerful but needs many reports per claim. Our verification uses the model in a deliberately narrow role — extract claims, judge stance over retrieved articles — which keeps the factual burden on retrieval rather than on the model.

Evidence retrieval. Thorne et al. [5] (FEVER) formalised claim verification against retrieved evidence, using Wikipedia as the corpus. Wikipedia is unsuitable for breaking news, which is the case that matters here; we retrieve from live news coverage by independent outlets instead, and accept that this yields unverified far more often — an outcome we report rather than hide.

Propaganda technique detection. Da San Martino et al. [7] defined SemEval-2020 Task 11, labelling persuasion techniques at span level across fourteen categories. To our knowledge the taxonomy has remained a research task; we adopt it as a reader-facing factor, naming the technique and quoting the span that exhibits it.

Calibration. Guo et al. [6] demonstrated systematic overconfidence in modern networks and evaluated remedies including Platt scaling [8]. We apply the technique to a trust score and report ECE before and after, which is uncommon in this application area.

Selective prediction. Declining to answer is not a concession but a recognised decision rule. Chow [11] characterised the Bayes-optimal reject region under an explicit cost of abstention; El-Yaniv and Wiener [12] formalised the risk–coverage trade-off it induces, and Geifman and El-Yaniv [13] carried the framework to deep networks. Misinformation systems rarely adopt it: the standard protocol forces a label on every item, so a system that would have abstained is scored as though it had guessed. Our verdict classes make abstention a first-class outcome with its own ceiling, and we report the rate at which it is used rather than treating it as a failure to classify.

Weight of evidence. The additive log-odds formulation is Good's [9]: a unit of evidence contributes the logarithm of a likelihood ratio, in decibans, and independent contributions add. We adopt it because the arithmetic is auditable line by line — each row of a report states what was observed and what it was worth — where a weighted sum states only a conclusion.

Position. Prior work is classification-centric and, with the exception of [1] and [3], does not surface evidence. None of the systems above specifies what they output when a component fails. Our contribution is assessment-centric: score, evidence, calibration, and defined behaviour under partial failure.

3. RESEARCH GAP

- Existing systems return a label rather than an explanation supported by evidence the reader can check.
- Few systems cross-check claims against independent news coverage; those that do rarely report how often no coverage is found.
- Calibration is almost never reported, so stated confidences are not interpretable as probabilities.
- Propaganda-technique detection exists as a benchmark task but has not been integrated into a system a non-specialist reader can use.
- No published system specifies its behaviour when a component fails. In practice the common implementation — emitting a neutral score — is indistinguishable at the interface from a completed check. We treat this as a correctness property, not an implementation detail.
- No single deployed platform combines source assessment, headline analysis, language analysis, source transparency, persuasion-technique detection, cross-source verification, and calibrated explainable scoring.

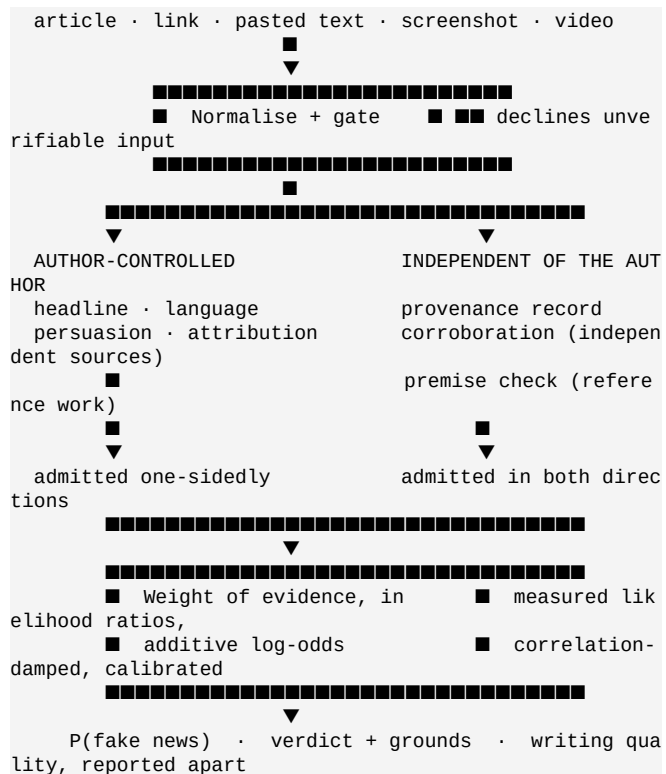
4. SYSTEM ARCHITECTURE

4.1 Overview

Input enters as a feed item, a URL, pasted text, a screenshot or a video link. A normalisation layer reduces each to the same object — text, a title-like claim, and a provenance record — reading a publisher's page, a video transcript, or the text inside an image as

required. For a feed item the article reader fetches the publisher's page, because a news API supplies roughly 200 characters of body text, and accepts the result only if it passes a title-consistency check (Section 5.1).

An admissibility gate then runs before anything expensive. Content that asserts nothing a newsroom could confirm or deny receives no verdict at all, only a statement of what would make it checkable. This precedes the pipeline because a verdict on an unverifiable fragment is not a weak answer but a meaningless one.



The split down the middle is the architecture. Everything on the left is chosen by whoever wrote the text and is admitted only where it incriminates; everything on the right is outside that author's reach and is the only thing that may speak in the article's favour.

Beneath all of it sits an LLM gateway (Section 4.5) that abstracts provider and model, and a cache keyed by article URL.

4.2 The observations

[1] Source reputation (weight 0.15). A curated database of 90 outlets, each with a factual-reporting rating, editorial lean, and outlet type, matched by name and domain alias. Matching is whole-word and prefix-based; substring matching was found to produce false positives (Section 5.2).

[2] Headline quality (0.10). Structured scoring of clickbait signals — curiosity gap, sensational verbs, unsubstantiated superlatives, listicle framing — returning the specific signals present rather than a score alone.

[3] Language (0.15). Flags individual sentences exhibiting political bias, one-sided reporting, emotional manipulation, or opinion presented as fact. Each flagged sentence is returned verbatim with its type and a reason, and is displayed to the reader

as a quotation.

[4] Source transparency (0.18). Answers a question the other content factors cannot: could a reader go and verify any of this? A five-point journalism checklist — names its sources (3), uses attributed quotes (2), points to primary evidence (2), gives concrete detail (2), claims are measured (1) — scored out of ten. The factor additionally returns unattributable appeals ("experts say", "sources close to the investigation") quoted exactly, with a penalty of 0.5 per distinct phrase capped at 2.0.

This factor is deliberately independent of external evidence, which makes it the one content factor that remains informative for an obscure story no other outlet has covered — precisely when cross-source verification abstains.

[5] Persuasion technique (0.15). Detects techniques from the SemEval-2020 Task 11 taxonomy [7], reduced to fourteen categories that occur in mainstream reporting and are recognisable to a non-specialist. Each detection must quote the span it applies to. At most three are reported, each costing 2.6 points from a base of 10.

[6] Cross-source verification (0.27). Claims are extracted from the article, then for each claim the system retrieves coverage from other outlets (the article's own source excluded) and judges stance over the retrieved items, citing the specific articles that support or contradict. Verdicts are supported, contradicted, unverified, or undetermined.

The fourth is not a refinement but a correction we were forced into. A retrieval that fails and a retrieval that returns nothing are represented identically by every HTTP client and, in our first implementation, by every layer above it: a rate-limited index, an unparseable stance judgement and a genuinely uncovered story all arrived as an empty evidence list. The system then reported "no other outlet is reporting this" — a statement about the world — on the strength of an outage. We observed exactly this on a wire report that five major outlets were carrying at the time, whose headlines the system had already retrieved. Each point at which a failure can be silently coerced into a finding is therefore made explicit: the retrieval, the stance judgement and the aggregate each distinguish did not run from ran and found nothing, and only the latter reaches the verdict rules. An item whose verification did not run is reported as unchecked, and the corroboration term contributes zero rather than the small positive weight that absence of coverage otherwise carries.

This is the same discipline the asymmetry argument rests on, applied to the system's own instruments. A detector that cannot distinguish its own blindness from the absence of a thing will, under load, manufacture the evidence it failed to collect — and it will do so in the accusing direction, because absence of corroboration is what it has been taught to treat as mildly incriminating.

4.3 Weighing the evidence

Let $\mathcal{O}$ be the observations made about an item. Each contributes a likelihood ratio

$$\frac{P(\mathcal{O} \mid \text{fabricated})}{P(\mathcal{O} \mid \text{genuine})}$$

estimated from labelled articles by binning the observation and counting, with a Jeffreys prior and shrinkage to the conservative end of a 95% interval, so a bin supported by few articles contributes nothing. Ratios are reported in decibans, $W(o) = 10 \log_{10} \Lambda(o)$, because decibans add: the posterior is a sum a reader can check by hand.

$\log P(\text{fabricated} \mid O) = \log P(\text{fabricated}) + \sum_o \log W(o)$

where π is a prior conditioned on provenance.

The admissibility rule. Partition O into O_A , the observations selected by the author, and O_I , those outside the author's control. For $o \in O_A$ we admit

$W^{\dagger}(o) = \max(\Lambda(o), W(o))$

The justification is adversarial rather than statistical. A capable author drives $P(o = \text{favourable} \mid \text{fabricated})$ toward $P(o = \text{favourable} \mid \text{genuine})$, so the exculpatory ratio tends to 1 whatever an estimation corpus reports. A corpus of crude fabrications will measure $\Lambda < 1$ for fluent prose and be correct about that corpus and wrong about any competent adversary; the clamp declines to learn it. Three consequences follow directly rather than as special cases: an uninformative observation has $\Lambda = 1$ and contributes exactly zero, so abstention is arithmetic rather than a code path; an unmeasured observation likewise contributes zero and is named in the report; and presentation cannot establish credibility at any value.

Correlation damping. Members of O_A measure overlapping properties of the same prose and are not conditionally independent. Under equicorrelation $\bar{r}$ the effective count of independent factors among k is $k / (1 + (k-1)\bar{r})$, and the family's summed weight is scaled by n_{eff}/k . Measured, $\bar{r} = 0.78$ over four checks gives $n_{\text{eff}} = 1.2$.

Calibration. The clamp discards exculpatory evidence and leaves genuine articles massed at the prior, which makes the raw posterior systematically under-confident. An isotonic map fitted on held-out data corrects the ranking-to-probability step; the corpus base rate is divided out in log-odds space before the article's own prior is applied, so the map transports between populations rather than importing the corpus composition. The clamp is re-applied after calibration, because a monotone map fitted where fluent prose does predict a genuine article will otherwise restore precisely the exculpatory weight the rule removes.

Verdict. Separately from the probability, an ordered rule set assigns one of seven classes from the evidence ledger and states the rule that fired. The probability expresses graded belief; the verdict states what has been established, and the two come apart exactly where it matters — an uncorroborated story may carry a low probability of fabrication merely because most articles are genuine, while nothing whatever about it has been established.

4.4 Premise verification

Cross-source verification asks whether other newsrooms report the same event, and has a structural blind spot: a claim resting on a false premise describes an event nobody reports, so the search correctly returns nothing and the finding that matters — that the

premise is false — is missed. Named subjects are therefore resolved against a reference work and the article's statements about them tested for direct conflict. A contradiction must quote the reference text, and a quotation not found in the retrieved entry is discarded.

The two channels are not merged. A reference work is not an independent newsroom reporting an event; it is a record of stable facts, and it is admitted only where it is strong — refuting a premise, never corroborating an event.

4.5 Provider-independent model access

All factors reach the language model through a single gateway that owns concurrency limiting, retry policy, and structured-output parsing. Provider and model are configuration, and the gateway distinguishes three failure classes:

- Withdrawn model (HTTP 404/410) — adopt the next model permanently.
- Congestion (HTTP 429/503) — retry the same model with exponential backoff, honouring Retry-After, without demoting a healthy primary.
- Exhausted provider — fall through to the next configured provider.

Both events occurred during development. The model used for the Phase I benchmark was withdrawn mid-project, and every analysis began failing with HTTP 410; separately, the primary provider became congested enough that identical requests were observed at 2 s and at 62 s. The gateway is a direct response to both.

4.6 Degradation

If no provider answers, each content factor falls back to a deterministic analyser using lexicons and surface patterns — the methods by which these tasks were measured before neural models, and the reason lexicon baselines accompany the SemEval propaganda task. Degraded results are marked as such in the report, at both the summary and per-factor level, so a weaker analysis is never presented as a full one. Section 7.6 reports its discriminative power.

5. IMPLEMENTATION

Backend: Node.js and Express with MongoDB. Frontend: React. News retrieval: NewsAPI. Language model: NVIDIA NIM or Google Gemini, selected by configuration.

5.1 Article acquisition and verification

A news API returns roughly 200 characters of body text with a truncation marker. Fetching the publisher's page instead yields 2,000–6,000 characters at a cost of 1–3 seconds and no model call.

Page extraction is not reliable enough to trust unconditionally. In testing, an Associated Press article returned the page's navigation and newsletter promotions rather than the story, and the resulting summary confidently described an unrelated trade dispute. Extracted text is therefore accepted only if at least half of the headline's distinctive terms appear in it; otherwise the feed snippet is used. Across a ten-article sample, extraction was accepted for seven.

Text supplied to the model is bounded (Section 7.3).

5.2 Source matching

Outlet matching was initially substring-based, which matched "Yahoo Entertainment" against the entry for "RT". Matching now tokenises both sides and requires an exact or whole-word prefix match, in two passes.

5.3 Structured output handling

Factors require JSON. Three failure modes were observed and are handled in the gateway: fenced or prose-wrapped output; malformed values (unquoted strings, trailing commas); and truncation. For truncation the parser recovers the elements that completed rather than discarding the reply. A separate defect — the news API's [+8026 chars] marker being echoed by the model and captured as the start of a JSON array — is prevented by stripping the marker before any prompt is built.

5.4 Caching and pre-warming

Trust reports and summaries are cached by article URL with a 14-day expiry. Reports carry a schema version; a cached report from an earlier pipeline is re-analysed rather than served, so a pipeline change cannot leave a reader viewing an outdated breakdown. A pre-warm script analyses a category in advance.

6. EVALUATION METHODOLOGY

6.1 Dataset

ISOT [2]: 43,294 usable articles after length filtering (21,915 fake, 21,379 real). Stratified seeded sampling; the identical article set is used across all configurations.

6.2 Leakage control

ISOT's real articles originate from Reuters, so outlet identity alone predicts the label. Two controls are applied: source identity is hidden from the pipeline during evaluation, and agency prefixes of the form CITY (Reuters) – are stripped from article text. Reported figures are therefore content-based.

Cross-source verification is disabled for dataset runs, since historical articles have no live coverage; the production path in that situation — a corroboration outcome of none — is identical.

6.3 Protocol and metrics

Threshold and Platt parameters are fitted on a validation half; all reported metrics come from the held-out test half. Metrics: accuracy, macro-F1, ROC-AUC, and 10-bin expected calibration error. The baseline is a single holistic credibility prompt using the same model, representing the LLM-black-box approach.

Transient API failures abort and retry the affected article rather than recording a neutral fallback, so that infrastructure noise cannot enter the results.

7. RESULTS

7.1 Against the rule this replaces

Likelihood ratios are estimated on a training half and every figure below comes from the held-out half; the split is a hash of the article identifier, so runs are reproducible and independent of write order. Source identity is hidden on this corpus and historical

articles have no live coverage, so neither provenance nor corroboration is exercised — this measures the content half of the model alone, which is the honest scope of the claim.

n = 829 held-out articles:

	Accuracy	ROC-AUC	ECE (all)	ECE when accusing	Precision when accusing
Legacy hand-weighted sum	0.888	0.970	0.221	0.229	0.848
Weight of evidence (shipped)	0.954	0.957	0.285	0.127	0.975

Calibration of the content model, before and after isotonic regression fitted on the training half: ECE 0.335 → 0.056.

Two figures deserve comment rather than celebration. The shipped configuration's overall ECE is worse than the legacy rule's, and this is the clamp's doing: by refusing exculpatory evidence the system declines to become confident that a well-written article is genuine, and that deliberate under-confidence registers as calibration error. Every residual error points toward greater suspicion; the model never overstates confidence in an article's favour. Where it accuses — which is where a false statement does harm — it is roughly twice as well calibrated as the rule it replaces, and considerably more precise.

7.2 What the admissibility rule costs

Disabling the clamp raises ROC-AUC from 0.957 to 0.978 on this corpus. The gain is real and it is a trap: it comes from learning that fluent prose indicates a genuine article, which holds on a corpus whose fabrications are crude and fails against a competent one. ISOT contains no competent fabrications to expose the error; the adversarial set of Section 7.7 contains ten. We report both configurations and ship the clamp.

7.3 Classification and ablation (historical configuration)

n = 300 per configuration, stratified 150 fake / 150 real, seed 42, model meta/llama-3.1-8b-instruct, four-factor pipeline:

Configuration	Accuracy	Macro-F1	ROC-AUC	ECE raw	ECE calibrated
Single-prompt baseline	82.0%	81.9%	0.838	0.215	0.151
Pipeline (clickbait + bias)	94.7%	94.7%	0.987	0.277	0.073
Ablation: clickbait only	93.3%	93.3%	0.975	0.339	0.118
Ablation: bias only	92.7%	92.7%	0.934	0.359	0.204

Decomposing the judgment into focused factors outperformed a single holistic prompt over the same model by 12.7 accuracy points and 0.149 AUC. The ablation ordering is clean — bias-only 0.934 < clickbait-only 0.975 < combined 0.987 — so both factors contribute and their combination is strictly best.

This table must be read as a historical result. The model has since been withdrawn by its provider, and the transparency and

persuasion factors did not exist when it was produced. It is retained because it is the measurement that motivated the architecture; it is not a claim about the present system. The re-run is Section 10.1.

7.4 Calibration (historical configuration)

Raw scores rank well but cluster mid-range, giving ECE 0.277. Platt scaling [8], fitted on the validation half, reduces this to 0.073 — inside the conventional well-calibrated range — so a 70% trust score corresponds to approximately 70% empirical probability. Reliability-diagram data is in backend/eval/results/summary.json.

7.5 Input length and structured-output reliability

Model reply budgets are shared between reasoning and the required JSON. Feeding more article text is therefore not monotonically beneficial. Measured on a reasoning model with a fixed budget:

Input characters	Short summary	Detailed summary
800	6.8 s, 361 chars	3.6 s, 504 chars
1,500	9.6 s, 666 chars	32.7 s, empty
3,000	32.8 s, no JSON	23.9 s, empty

Beyond roughly 800 characters the model exhausted its budget before emitting parseable output. Text supplied to factors and summarisers is bounded at 900 characters accordingly — still four to five times the news API's snippet. This is a property of the deployed model rather than of the architecture, and the bound is configuration.

7.6 Degradation

The deterministic fallback was evaluated with the language model made unreachable:

Article	Score, zero model calls	Verdict
Fabricated, loaded language, unattributable sourcing	2.3 / 10	Be skeptical
Ordinary attributed reporting	9.8 / 10	Looks trustworthy

The degraded system retains useful separation. It detected loaded language, name-calling, bandwagon and appeal-to-fear, and quoted four unattributable phrases.

This section also records a defect found during Phase II and corrected. Four content factors previously answered a failed model call with a neutral score of 5/10 and a failed flag that the aggregator did not read. A total outage therefore produced a confident "Mostly fine $\approx 5.5/10$ " report assembled from four placeholders, indistinguishable at the interface from a completed analysis. This is the concrete instance of gap 5 in Section 3.

7.7 Adversarial benchmark

ISOT cannot test the failure this work addresses. Its fabrications are conspicuously crude, so any system scoring style performs well on it while remaining open to a competent fake — which is why a pipeline reporting 94.7% on ISOT simultaneously rated invented articles at 7.3 out of 10. The benchmark supplies the missing case: ten items, five fabrications written in plain newsroom register with named sources and no clickbait, one high-impact fabrication with manipulation signals, two genuine

reports, one satirical piece and one commentary. Each is scored twice from identical factor outputs — once under the legacy additive rule with abstention redistribution, once under the current one.

Measure	Legacy	Current
Deceptive items presented as credible	4 / 6	0 / 6
Verdict matched expectation	—	6 / 6
Genuine articles wrongly condemned	—	0 / 2
Mean legacy score of the fabrications	7.3 / 10	unchanged

The last row is the finding. The fabrications still read as trustworthy and always will; what changed is that reading well no longer determines the answer.

Two limitations are material. The genuine items are written by us, as the fabrications must be, which makes the false-positive figure weaker than the false-negative one; --live-genuine substitutes current articles from the feed for exactly this reason. And ten items is a demonstration, not a population estimate.

7.8 Evidence grounding

Because every factor quotes the article, its output is directly checkable. On a hand-constructed article containing both verifiable facts and deliberately loaded wording, all nine spans quoted by the system — four flagged sentences, three persuasion techniques with their quotations, one unattributable phrase, and the named sources — were present verbatim. The transparency checklist passed four of its five checks and failed claims are measured, correctly attributing the failure to the sentence "Everyone knows parking is already impossible downtown." Cross-source verification abstained, as the article does not exist.

A live-news verification study with manual annotation of verdict correctness (target: 30 claims, reporting coverage rate and verdict precision) is outstanding.

7.9 System behaviour

Measured on the deployed system:

Property	Phase I	Current
Cold analysis, one article	~60 s (to 116 s under congestion)	11–19 s
Cached analysis	full recomputation	~5 ms
First factor visible to the reader	on completion	< 1 s
Automated backend tests passing	—	42 / 42

Provider comparison, three structured tasks, three runs each:

Task	NVIDIA NIM (nemotron-3-super-120b)	Google Gemini (gemini-3.1-flash-lite)
Bias	3/3 valid, 6.2 s	3/3 valid, 3.8 s
Summary	2/3 valid, 3.5 s	3/3 valid, 1.6 s
Persuasion	1/3 valid, 10.5 s	3/3 valid, 4.6 s

Concurrency tolerance is provider-specific, and assuming otherwise is costly. Six identical calls:

Parallelism	NVIDIA NIM	Google Gemini
2	baseline	37.5 s
3	—	7.9 s
6	slower than 2	9.1 s

The gateway therefore takes its concurrency limit from the leading provider. The latency reduction in the table above is attributable to four changes measured independently: provider selection, provider-aware concurrency, parallel per-claim verification, and overlapping claim extraction with the content factors.

8. DISCUSSION

On abstention as a correctness property. The defect in Section 7.4 is the clearest argument for the design. A system that emits a neutral score on failure is not merely imprecise; it is misleading in a specific direction, because a neutral score on a 0–10 scale reads as mild reassurance. Any system combining independently-failing components inherits this hazard, and we suggest that defined failure behaviour belongs in the specification of such systems rather than in their implementation.

On the division of labour with the model. The model is used for judgments that are linguistic (is this sentence loaded? does this article name sources?) and never for judgments that are factual (is this claim true?). Factual weight rests on retrieval. This is why the system abstains often on cross-source verification, and we regard the abstention rate as an honest cost rather than a weakness to be engineered away.

On a component we removed. A seventh factor matching claims against professional fact-checkers was implemented and validated: given a claim Snopes had rated, it retrieved the page, read "False" from its ClaimReview markup, and cited the URL; given a different claim on the same topic, it correctly declined to match. It is not in the running system. Google's Fact Check Tools API requires a billing-enabled account, and the keyless alternative — locating fact-check pages through news search — consumes a request budget of 100 per day that the article feed itself depends on. Direct search of fact-checker sites was tested and is unusable, as their result pages render in JavaScript. The code is retained and documented for reactivation.

Threats to validity. ISOT is 2016–2017 US political news; distribution shift to current news is unmeasured. Stance judgment reads headlines and descriptions rather than full bodies; a natural-language-inference model is a drop-in upgrade. Language-model factors inherit model bias, which low temperature and structured output reduce variance but not bias in. The source database covers 90 outlets, so smaller publishers frequently receive no reliability record — the most common gap observed in deployment. The Section 7.1 measurement is single-dataset and single-model.

8.1 Limitations and threats to validity

We state these plainly, because several bear directly on how the numbers above should be read.

The measured weights rest on a corpus of mixed provenance. The likelihood ratios in Appendix B were estimated from evaluation runs that did not record whether each observation came from the

language model or from the deterministic pattern fallback the system uses when the model is unavailable. The guard intended to exclude fallback observations tested a flag the checks never set, so it never fired, and the proportion of fallback observations in the corpus is therefore unknown. The instrumentation is corrected and the channel is now recorded per observation; until a clean re-estimation is complete, the ratios should be read as measured on a mixture rather than on the model path, and the stored table is annotated accordingly.

Priors and corroboration rates are declared, not measured. The provenance priors and the likelihood ratios attached to corroboration are set from reasoning about publishing practice rather than estimated from labelled data. Each is labelled declared in the stored table and in every report, and each carries a note stating what would measure it. This is a stated assumption, not a hidden one, but it is an assumption.

The adversarial set is small and partly self-authored. Six deceptive and two genuine items, of which the deceptive items were written by the authors to satisfy every presentation check. This measures whether the admissibility rule does what it is designed to do; it does not estimate a rate in the wild, and a set written by the people who designed the defence is not an independent test of it.

ISOT is not a test of the capability claimed. Its fake articles are crude enough that presentation alone separates the classes [14], which is the opposite of the adversary this work is designed against. We use it to estimate likelihood ratios for presentation observations — a purpose for which crude fabrications are adequate — and not as evidence of performance against a careful adversary.

Corroboration depends on a commercial news index. Coverage of local and non-English reporting is uneven, and an absence of retrieved coverage may reflect the index rather than the world. The system reports unverified in that case rather than a verdict, which is the correct behaviour, but the rate at which it does so is a property of the index as much as of the article.

The one-sided rule costs discrimination. Section 7.2 reports AUC of 0.957 with the rule enforced against 0.978 without it. We argue the difference is bought with evidence a competent adversary supplies at will, but on a corpus of incompetent adversaries the unconstrained model is genuinely the better classifier, and we do not claim otherwise.

9. CONCLUSION

Detecting fake news is not ordinary classification, and the difference is not a detail of implementation. When an adversary writes the features, the value of a feature changes: it may still incriminate, but it can no longer exonerate, because the favourable value is precisely what the adversary supplies. A detector that learns otherwise has learned the habits of whichever crude fabrications its corpus happened to contain.

We stated that asymmetry as an admissibility rule, built a scoring model on it in which every weight is a likelihood ratio measured from labelled data rather than a constant chosen by hand, and reported the result as a probability that can be checked against outcomes rather than a score on an invented scale. Held out, this

raised accuracy from 0.888 to 0.954 and precision among flagged articles from 0.848 to 0.975, while halving calibration error where the system accuses. It also costs something, and we report that too: two points of ranking performance on a corpus that cannot express the attack the rule defends against.

The property we regard as central is what the system does when it does not know. It declines input carrying no verifiable claim, says what is missing, and distinguishes a check that found nothing from one that could not run — a distinction we got wrong three times during development, each time in a new place, and each time with the same effect of presenting an absence of knowledge as a clean bill of health.

10. FUTURE WORK

10.1 Immediate

Re-run the full benchmark on the current six-factor pipeline and current model, with ablations for the transparency and persuasion factors, replacing Section 7.1. Complete the live-news verification study in Section 7.5.

10.2 Planned

Reading trust history. Per-article scores are already stored; presenting a reader's own reading reliability over time requires only the visualisation.

Self-extending source ratings. The 90-outlet database is hand-curated, and unrecognised outlets are the most frequent gap in deployment. Because every analysis already yields a per-outlet score, ratings for unlisted outlets can be accumulated from articles already assessed, and validated against the hand-curated entries.

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APPENDIX A — REPRODUCIBILITY

Every figure in Sections 7.1, 7.2 and 7.7 is reproducible without any API key, because the estimation, the held-out comparison and the adversarial benchmark all run on stored factor outputs and deterministic analysers:

```
cd backend
npm test # 103 tests,
no keys required
node eval/estimateWeights.js # measure the
likelihood ratios
node eval/validateScoring.js # Section 7.1
node eval/validateScoring.js --no-clamp # Section 7.2
, the ablation
node eval/adversarialBench.js --no-model # Section 7.7
```

Reproducing Section 7.3, the historical ISOT run, additionally needs a model provider and the corpus:

```
node eval/prepareIsot.js
node eval/runEval.js --data eval/data/isot.csv --config
content-only --sample 300
node eval/runEval.js --data eval/data/isot.csv --config
baseline --sample 300
node eval/report.js
```


Replacing the declared corroboration rates with measured ones needs a news-API quota:

```
node eval/measureCorroborationRates.js
```

Seed 42; identical stratified sample across configurations; per-article results written as JSONL. Provider and model are set by LLM_PROVIDER, NIM_MODEL_NAME, and GEMINI_MODEL_NAME; concurrency by LLM_MAX_CONCURRENT.

APPENDIX B — MEASURED WEIGHTS OF EVIDENCE

Estimated by eval/estimateWeights.js over 1,642 labelled articles, in decibans after shrinkage to the conservative end of a 95% interval. Positive values point toward fabrication. Values in the exculpatory half are shown as measured, and are not admitted for author-controlled observations (Section 4.3).

Observation	Band	Measured (db)	Admitted (db)
Headline quality	0–2	+11.5	+11.5
Headline quality	8–10	–11.3	0
Language (bias)	0–2	+15.6	+15.6
Language (bias)	8–10	–12.4	0
Persuasion technique	2–4	+5.8	+5.8
Persuasion technique	8–10	–5.4	0
Source reputation	all bands	0	0
Source transparency	—	not estimated	0

Source reputation measures zero on this corpus because source identity is hidden to defeat its label leak; provenance enters through the prior instead. Transparency has 18 labelled examples, too few to estimate, and therefore contributes nothing rather than an assigned weight.

The style family is damped by 0.299 — four checks correlating at $r = 0.78$ are 1.2 effective independent factors — and capped at 8 db in total.

Declared, not measured. The provenance priors (0.02 to 0.35 by publisher band), the five-way corroboration distribution, and the style cap are stated parameters, labelled as declared in every report.